ISSSP 2026 · Hong Kong Finals

BOBER Veins / VascuShift-GPINN

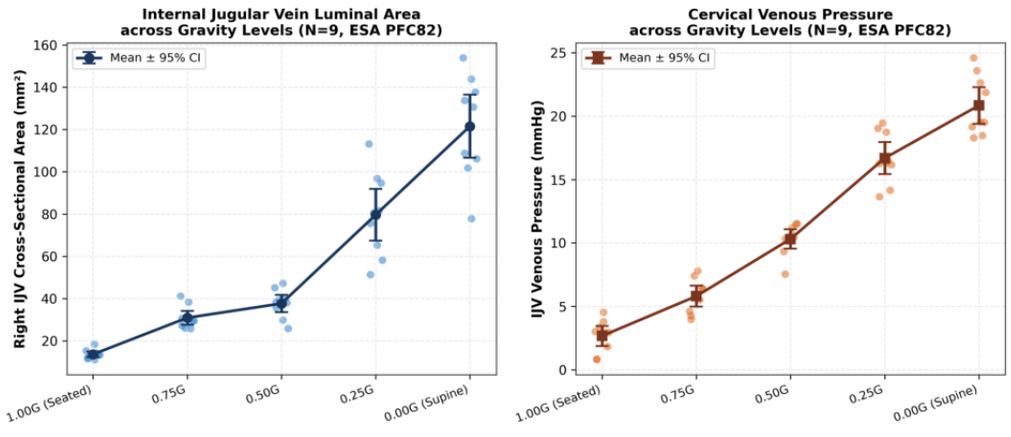
Physics-Informed Modeling of Gravity-Dependent Venous Hemodynamics & Optical rPPG

Authors: Daniel, Natalia, Michał, Jakub
Space Medicine Payload Group

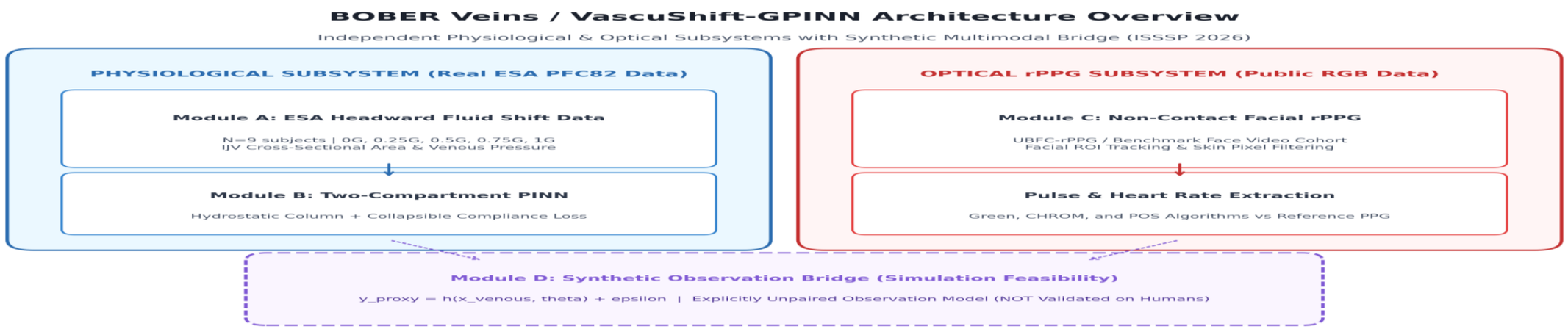
1. SCIENTIFIC MOTIVATION & PROBLEM

- Cephalad Fluid Shift: Upright 1G posture produces a vertical hydrostatic drop ΔPh = ρ·g·Δh ≈ 27.3 mmHg, keeping internal jugular veins (IJV) collapsed.
- Microgravity (0G): Eliminates this column (ΔPh = 0), driving severe jugular engorgement (+800% area) and elevated cervical venous pressure (21 mmHg).
- SANS & Thrombosis: Persistent venous congestion impairs ocular and cranial drainage, contributing to Spaceflight-Associated Neuro-ocular Syndrome and documented cases of jugular venous blood flow stasis in astronauts.
- Goal: Develop an interpretable physics-informed model and evaluate non-contact optical rPPG feasibility for autonomous spaceflight health monitoring.

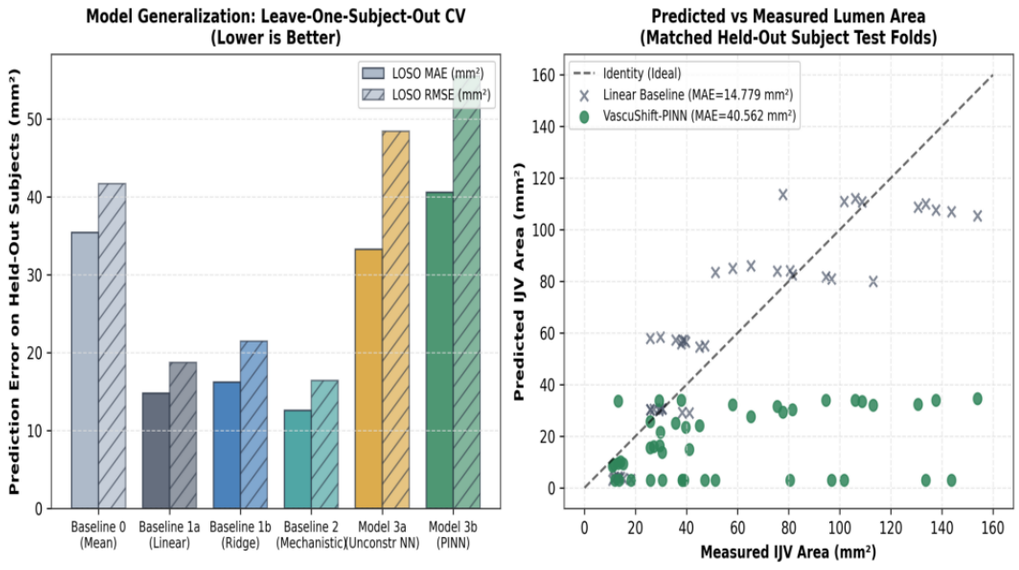
2. REAL DATA: ESA 82nd PARABOLIC FLIGHT



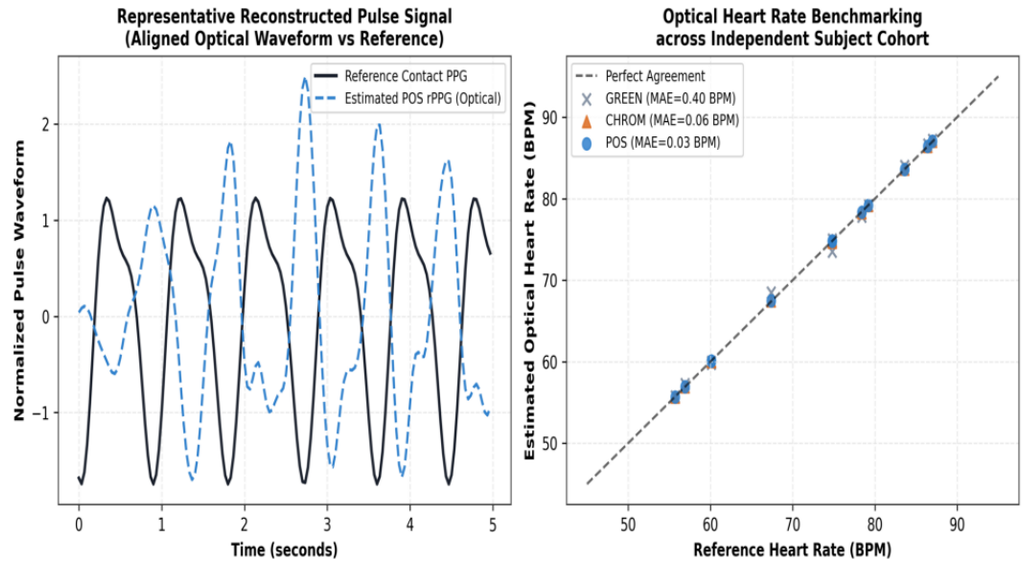
3. SYSTEM ARCHITECTURE & SCIENTIFIC BOUNDARIES



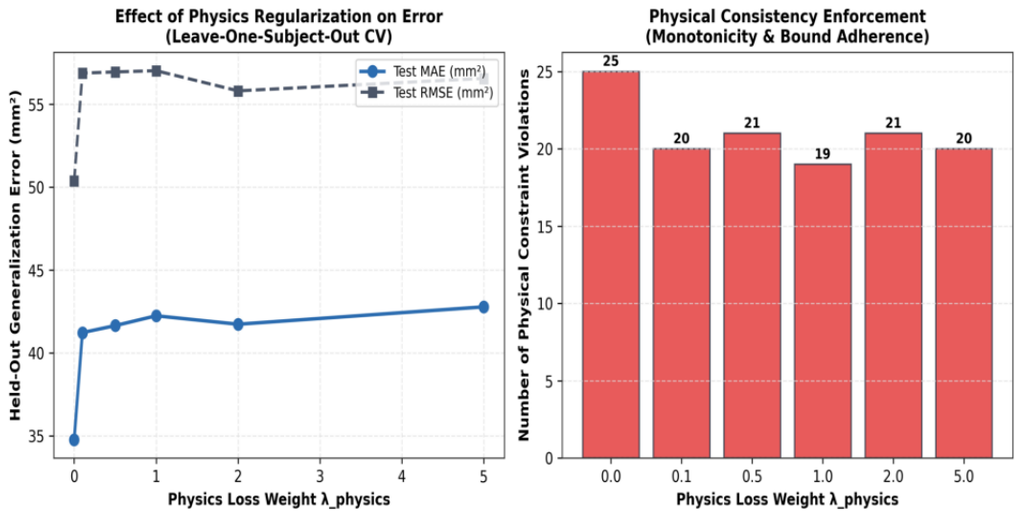
4. RESULTS: LEAVE-ONE-SUBJECT-OUT CV



5. RESULTS: OPTICAL rPPG BENCHMARK



6. IDENTIFIABILITY & PHYSICS ABLATIONS



7. DEMONSTRATOR & MISSION PAYLOAD

